100 DAYS OF MACHINE LEARNING

DAY 2

Complete Notes, Examples & Practice Workbook

Official CampusX playlist study guide • Videos 4–7

Topics: Batch vs Online Learning • Instance-Based vs Model-Based Learning • Challenges in Machine Learning

Study duration: approximately 71 minutes of video content, plus notes, review, and coding practice.

Assessment result: Video 7 quiz — 5/5 (100%)

Beginner-friendly explanations in English with Roman Urdu reminders

How to Use This Workbook

Read one section at a time. Explain each concept in your own words, then run the accompanying Python practice file. The examples are educational demonstrations, not production-ready systems.

Day 2 learning goals

• Explain batch (offline) learning and online learning.

• Distinguish the way a model is updated from the way it makes predictions.

• Compare instance-based learning with model-based learning.

• Recognize insufficient data, non-representative data, poor-quality data, irrelevant features, overfitting, underfitting, and data drift.

• Use a small Python example to see nearest-neighbor and linear-regression ideas.

Video Roadmap

Video	Topic	Approx. duration	Main question
4	Batch / Offline Learning	11 min	When does the model learn from new data?
5	Online Learning	19 min	How can the model update incrementally?
6	Instance-Based vs Model-Based	17 min	Does prediction rely on stored examples or a learned model?
7	Challenges in Machine Learning	24 min	Why can a model fail even when we have an algorithm?

Approximate total: 71 minutes. Video durations can vary slightly by playback version.

1. Batch (Offline) Learning

Definition: Batch learning trains a model using a collected dataset in a separate training process. Once training is complete, the model is used to make predictions. To incorporate new information, the team commonly prepares updated data and retrains the model.

Example: A university collects last semester's course, attendance, and assessment data. The data team trains a model, evaluates it, and deploys it. At the end of the next semester, it retrains using the newer data.

Advantages

• Training can be planned and tested before deployment.

• The deployed model may be stable between retraining cycles.

• It can be practical when data arrives in periodic batches.

Limitations

• New information may not affect the model until the next retraining cycle.

• Retraining can require time and computing resources.

• A model can become stale if the real-world pattern changes quickly.

Roman Urdu: Batch learning mein pehle data jama hota hai, phir model ko aik training cycle mein train kiya jata hai. Naya data shamil karne ke liye aksar dobara training karni parti hai.

2. Online Learning

Definition: Online learning updates a model incrementally as new examples or small batches arrive. Depending on the system, updates can happen continuously or on a schedule. It is useful when data streams in or patterns change frequently.

Example: A service receives new transaction examples throughout the day and periodically updates its fraud model after suitable validation checks.

Advantages

• Can adapt to new data sooner than a long batch retraining cycle.

• May avoid retraining from scratch on the entire historical dataset.

• Useful for high-volume or continuously arriving data when the algorithm supports incremental updates.

Risks and safeguards

• Noisy or malicious data can damage the model if accepted blindly.

• Monitor metrics and data quality; keep rollback options.

• Use appropriate validation, learning-rate settings, and drift checks. Not every algorithm supports online updates.

Roman Urdu: Online learning mein model naye examples ya chhote batches se step-by-step update hota rehta hai. Lekin har naye data ko bina check kiye accept karna risky ho sakta hai.

3. Batch vs Online Learning

Dimension	Batch / Offline	Online / Incremental
Training pattern	Train on a collected dataset in a separate cycle	Update as new examples or mini-batches arrive
New data	Often waits for retraining	Can influence the model sooner
Best fit	Periodic updates and stable workflows	Streaming or rapidly changing data, if supported
Main concern	Model may become stale between updates	Bad data or unstable updates can harm performance
Example	Monthly demand-model retraining	Incremental updates from new events

Important: batch vs online describes the update/training schedule. It does not, by itself, tell you whether the model is instance-based or model-based.

4. Instance-Based Learning

Definition: Instance-based learning stores examples and predicts for a new example by comparing it with stored cases. A common example is k-nearest neighbors (KNN). It is often called a lazy-learning approach because much of the work happens at prediction time.

Simple example: Suppose a new student has 82% attendance and 74% in a prior assessment. A KNN-style method finds similar students in historical data and uses their outcomes to help predict a category or value. This is only an illustration; real use needs careful feature selection, privacy protection, and evaluation.

Typical strengths

• Easy to explain with the idea of similar examples.

• Can represent local patterns without fitting a complex global formula.

• Adding a new stored example may be straightforward.

Typical limitations

• Prediction can be slow when many examples must be searched.

• Storing examples can consume memory.

• Distance-based methods can be sensitive to feature scales and irrelevant features.

Roman Urdu: Instance-based model naye case ko purane milte-julte examples ke saath compare karta hai. KNN mein 'nearest neighbors' yani sab se qareebi examples dekhe jate hain.

5. Model-Based Learning

Definition: Model-based learning uses training data to fit a general pattern or set of parameters. It then uses the learned model to predict for new examples.

Example: Linear regression learns a relationship between house size and price from historical examples. After training, it uses the learned relationship to estimate the price of a house with a new size.

Typical strengths

• Prediction can be efficient after training.

• A compact model may use less memory than storing all training examples.

• Some models provide interpretable parameters or feature effects.

Typical limitations

• A model that is too simple may miss important patterns.

• Training choices and assumptions affect performance.

• The learned relationship may fail when the real-world data changes.

Roman Urdu: Model-based learning data se aik general pattern seekhta hai. Phir naye example par har purane record ko dhoondhne ke bajaye learned model prediction deta hai.

6. Instance-Based vs Model-Based

Question	Instance-Based	Model-Based
What is retained?	Many or selected training examples	Learned parameters or structure
How is a prediction made?	Compare new case with stored cases	Apply the learned model
Common example	K-nearest neighbors	Linear regression, decision tree
Common trade-off	More memory/search at prediction time	Training effort; model assumptions matter
Useful when	Similarity among examples is meaningful	A learned general relationship predicts well

These are different approaches to learning and prediction. Batch/online and instance/model-based are separate dimensions: a model can be trained in batches or incrementally, depending on the algorithm and implementation.

7. Challenges in Machine Learning

A model can use a valid algorithm and still fail because the data is inadequate, the features are weak, the model has learned the wrong level of complexity, or the real world has changed. Always evaluate on data that reflects the intended use.

Challenge	What it means	Example	Possible response
Insufficient data	Too few useful examples to learn a reliable pattern	Only 20 examples for a complex task	Collect more relevant examples; simplify the task/model; use suitable validation
Non-representative data	Training examples do not reflect the population where the model will be used	Only urban customers, but deployment includes rural customers	Sample relevant groups; check coverage and subgroup metrics
Poor-quality data	Missing, duplicated, inconsistent, mislabeled, or incorrect values	Exam marks entered as 900 instead of 90	Profile, validate, clean, and document data handling
Irrelevant features	Inputs contain little useful signal or distract the model	Predicting exam success using a random row ID	Use domain reasoning and validation to select useful features
Overfitting	Model learns training-specific details/noise and generalizes poorly	99% train accuracy, 60% test accuracy	Cross-validation, regularization, simpler model, more representative data
Underfitting	Model is too simple or not adequately trained to learn important patterns	Low accuracy on both training and test data	Improve features/model capacity; train appropriately; inspect preprocessing
Data drift / changing patterns	Input data or the relationship between inputs and outcomes changes over time	Customer preferences change, recommendations worsen	Monitor input and outcome metrics; investigate and retrain when justified

8. Overfitting vs Underfitting

Clue	Overfitting	Underfitting
Training performance	Often very good	Often poor
Test / unseen performance	Much worse than training	Also poor
Likely explanation	Learns noise or training-specific details	Fails to capture important structure
Possible remedies	Simplify, regularize, cross-validate, improve data coverage	Improve features, increase suitable model capacity, train correctly

Remember: do not diagnose a model from training accuracy alone. Compare training and validation/test performance, use a suitable split, and check whether the test data reflects real deployment.

9. Practical Machine-Learning Workflow

Step	Action	Question to ask
1. Define	Specify prediction goal and success metric	What decision will this model support?
2. Collect	Gather relevant and permitted data	Does it cover the intended users and conditions?
3. Inspect	Check types, missing values, duplicates, labels, outliers	Are values plausible and consistent?

Step	Action	Question to ask
4. Prepare	Split data; clean; encode/scale where needed; prevent leakage	Did information from the test set leak into training?
5. Establish baseline	Train a simple reference model	Is the complex approach actually better?
6. Evaluate	Use validation/test metrics and error analysis	Which cases fail, and for whom?
7. Deploy carefully	Document limitations and monitor outcomes	Does real-world behavior match evaluation?
8. Maintain	Investigate drift; retrain only when justified	Has performance or data changed meaningfully?

10. Python Practice

The accompanying single Python file includes two small demonstrations: a nearest-neighbor example to illustrate instance-based prediction, and linear regression to illustrate model-based learning. It also contains a data-quality checklist and challenge scenarios.

Requirements: Python 3. If scikit-learn and NumPy are installed, run the full demonstrations. Installation command: `python -m pip install numpy scikit-learn`

Expected learning, not a guaranteed real-world score: KNN compares a new point with stored examples; linear regression learns a line from training examples. The examples use tiny synthetic datasets only.

11. Exercises

A. A model has high training accuracy but much lower validation accuracy. Name the likely challenge and one possible remedy.

B. You receive new data every second. What update approach might fit, and what safety check should remain in place?

C. Why should numerical features often be scaled before a distance-based KNN model?

D. A model performs poorly on both training and test data. What is the likely challenge?

E. A model's users change over time and performance falls. What should you monitor or investigate?

F. Explain in your own words why batch/online is a different distinction from instance/model-based.

12. Worked Answers

• **A.** Likely overfitting. Consider cross-validation, regularization, a simpler model, and more representative data.

• **B.** Online/incremental updates may fit if the algorithm supports them. Validate data quality, monitor performance, and retain rollback safeguards.

• **C.** Features measured on large scales can dominate distance calculations, even when they are not more important. Scaling can make distances more balanced.

• **D.** Likely underfitting; inspect features, model capacity, preprocessing, and training setup.

• **E.** Monitor data quality, input distributions, outcome metrics, and subgroup performance; investigate data or concept drift.

• **F.** Batch/online concerns when updates occur; instance/model-based concerns how the learning/prediction method uses examples or a fitted model.

13. Day 2 Video 7 Quiz Review

Result: 5/5 — 100%. All answers were correct.

Question clue	Correct concept	Reason
99% training, 60% unseen	Overfitting	Large train/test performance gap
Poor on train and test	Underfitting	The model has not learned useful patterns
Urban-only training; rural use	Non-representative data	Training sample does not cover deployment population
Missing and incorrect marks	Data quality / cleaning	Inspect and handle errors before training
Preferences change over time	Data drift / changing patterns	The data or relationship may no longer match training

14. Revision Checklist

■ I can explain batch learning and online learning without memorizing a sentence.

■ I can state that batch/online and instance/model-based describe different dimensions.

■ I can explain KNN as comparing a new example with nearby stored examples.

■ I can explain linear regression as learning a relationship from examples.

■ I can distinguish overfitting from underfitting using training and test performance.

■ I can identify non-representative data, poor data quality, irrelevant features, and data drift.

■ I can explain why real-world monitoring and careful evaluation matter.

Day 2 completion summary

Videos studied: 4–7 • Approximate video time: 71 minutes • Video 7 quiz: 5/5 (100%).

Before moving to Video 8, make sure you can explain the four major distinctions in your own words: batch vs online, instance-based vs model-based, overfitting vs underfitting, and representative vs non-representative data.

Source alignment: official CampusX 100 Days of Machine Learning playlist, in the supplied playlist order. This workbook is a learning aid, not a verbatim transcript.